Device Power Consumption & Solar Sizing Report (24h Analysis)

1. Daily Energy Consumption
Total: 7.613 Wh/day
Active: 7.333 Wh (~3.72 h)
Sleep: 0.280 Wh (~20.26 h)
Active dominates (~96.3% energy)

2. Power Profile
Active: ~1.97 W
Sleep: ~0.014 W (14 mW)

3. System Losses
Efficiency: 70%
Required: 10.9 Wh/day

4. Solar Sizing (Germany)
Sunny (5 PSH): 2.2 W
Average (3 PSH): 3.6 W
Cloudy (1.5 PSH): 7.3 W
Winter (1 PSH): 10.9 W

5. Recommendation
10–15 W solar panel
Battery: 15–25 Wh (2–3 days autonomy)

6. Conclusion
Works fine in theory. Winter decides reality.

Metric	Value
Total Energy	7.613 Wh/day
Active Energy	7.333 Wh
Sleep Energy	0.280 Wh
Efficiency	70%
Adjusted Energy	10.9 Wh/day
Recommended Panel	10–15 W

consumption is 7.6Wh/day
if we consider 70% efficiency then 10.9 or 11Wh/day
20W of solar panel will generate approx 17Wh/day during Bad Days
15W of solar panel will generate approx 12.75Wh/day during Bad Day